# Shyamali Naik

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https://github.com/shyamalinaik369-ops | https://shyamalinaik369-ops.github.io

Professional Summary

An aspiring data analyst master's student with a proven ability to develop and deploy machine learning models to solve complex financial, environmental, and business problems. Transitioned from an engineering background, bringing a rigorous, quantitative approach to data-driven decision-making. Skilled in Python, SQL, R, and Tableau, with a standout thesis project that generated a **10.87% annualized alpha with Maximum Drawdown of -17.14%** in a momentum-based trading strategy. Eager to apply technical depth and a passion for uncovering insights to a challenging Data Analyst role.

## Technical Skills

| Category | Skills |
|---|---|
| **Programming & Databases** | Python, R, SQL, JAVA(Basic), MATLAB, C |
| **Data Science & ML Libraries** | Pandas, NumPy, Scikit-Learn, Matplotlib, Seaborn, PyTorch, LightGBM, QuantStats |
| **Data Visualization & BI** | Tableau, Power BI |
| **Big Data Technologies (Self-Study)** | Hadoop, Spark, Hive (Familiarity with core concepts) |
| **Developer Tools** | Git, Jupyter Notebook, Advance Excel (Pivot Tables, VLOOKUP, Formulas), AWS (Basic) |

## Education

**MSc Data Analytics | Queen Mary University of London** (September 2024 - December 2025)

- ***Relevant Modules***: Probability & Statistics, Machine Learning with python, Advanced ML, Neural Networks, Time Series Analysis, Financial Analytics (Python, R, SQL, Excel), Data Storing Manipulation & Visualization.
- ***Thesis***: Mitigating Momentum Crashes: An Adaptive, regime-Aware Strategy for Indian Equity Market.

**BTech in Civil Engineering | Gandhi Institute of Engineering & Technology University, Odisha, India** (2016 – 2020)

- ***Final Year Project***: Seismic Data Analysis using MATLAB & Statistical Modelling, Applied Remote sensing and GIS techniques to map ground water prospects, Facilitating water resource management.
- ***Research Projects***
- ➢ Assessing Tropical Cyclone Risk and Mitigation Strategies in Coastal Odisha, India: A Statistical Approach Using MATLAB.
- ➢ Statistical Analysis and Python-Based Optimization of Cyclone-Resistant Building Design.

## Thesis and Projects (MSc Data Analytics)

**Factor Investing: A Momentum Based Strategy | Final Thesis**

**Mitigating Momentum Crashes: An Adaptive, Regime-Aware Strategy for the Indian Equity Market**

- Designed a robust momentum investing strategy using historical stock market data from **NSE (2015-2024)**, including portfolio simulation and risk return profiling.
- Roadmap includes baseline momentum strategy, sector rotation models, and advanced ML layers (XGBoost, Clustering, SHAP analysis).
- Then explored **regime-aware strategy adjustments** (e.g., during market crashes) and potential integration of global comparison and alternative data (Google trends, Sentiment) and achieved a **10.87% annualized Alpha** by this strategy, which reduced the **Maximum Drawdown from -43.92 to -17.14%.**

**Bank Customer Churn Prediction | Classification Modelling Project**

- Improved Churn identification by outperforming baseline models, achieving a **63% recall and** 0.71 AUC with a SMOTE-Enhanced logistic regression model built from scratch to understand core ML logic.
- Demonstrated that recall is more critical than accuracy for identifying at risk customers in imbalanced datasets. Recommended targeted retention strategies based on churn prediction insights.

**Agro-Food Emission Analysis | Environmental Data Science Project**

- Built a custom database and performed data wrangling on global food production and emissions using Pandas. Conducted EDA across variables like CO2 trends, food emissions, global temperature rise, forest/fire impact, and population growth.
- Created compelling visualizations with Matplotlib and Seaborn to illustrate country wise and continent-wise emission pattern with a deep-dive on the Russian Federation (1990-2020). Delivered insights on the link between agri-food systems and climate trends, demonstrating analytical storytelling and environmental data.

**Forecasting U.S. Consumption | Time Series Analysis**

- Modeled quarterly changes in U.S Consumption using multivariate ARIMA model**s** in R, incorporating variables like income, production, and unemployment.
- Evaluated model performance using RMSE, MAPE, Theil's U, and residual diagnostics (ACF, Ljung-Box Test). Selected MODEL 1 (ARIMA with income + production) for its balance of interpretability, statistical validity, and white noise residuals, despite other models showing lower RMSE but failing logistic checks.
- Demonstrated that simplicity and residual robustness often outperform overfitting in economic forecasting models.

**Autoencoders on MNIST | Neural Networks & Latent Space Analysis**

- Built a deep autoencoder to learn compressed representations of handwritten digits and performed advance latent space analysis. Visualized bottleneck activation using t-SNE, explored neuron specialization and latent space interpolation to evaluate encoding depth.
- Conducted **neuron sensitivity analysis and visualized digit reconstructions by maximizing neuron activations**. Identified limitations in latent interpretability, motivating future work on disentangled representations and interpretable feature learning.

**Advanced Machine Learning: Galaxy Clustering, Image Completion and Reconstruction**

- Applied K-Means clustering to identify distinct galaxy clusters, using PCA for dimensionality reduction and 3D visualization on MNIST dataset. Implemented a matrix completion algorithm to successfully reconstruct missing pixels of cat image.

## Work Experience

**Engineering Intern -Structural Analysis & Quantitative Modelling PWD (Public Works Department) Odisha, India** (Feb 2021 – Dec 2022)

- Performed quantitative analysis and data validation for public infrastructure projects, ensuring precision in design specifications for road networks and bridges.
- Collaborated in a multi-disciplinary team of 10+ engineers, contributing **to data-driven quality checks** against rigorous safety standards and engineering codes.
- Utilized engineering software (AutoCAD, Revit Structure, ETABs, CSI Bridge, STAAD. Pro, SAP200) for **complex modelling and simulation tasks**, developing a disciplined approach to handling high stakes, real-world construction requirements.